Ahmed Gamal Elnopy

BIM Electrical Engineer

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Mail| LinkedIn Profile | Abu Dhabi, UAE

Profile

BIM Electrical Engineer with 6 years of experience in electrical design and technical office engineering on commercial, governmental and entertainment projects in the UAE, Saudi Arabia and Egypt. Skilled in LV/MV design, including load schedules, single-line diagrams (SLDs), cable sizing, voltage drop, short-circuit and lighting calculations, and bill of quantities (BOQ) preparation, as well as light current / ELV systems design (fire alarm to NFPA 72, data networks, CCTV and sound systems). Experienced in shop drawings, issued for construction (IFC) drawings, material submittals and requests for information (RFIs) to ADDC/AADC EWR requirements, and in MEP coordination and clash detection using Revit and Navisworks, with automated workflows built in Dynamo and C#.

Work Experience

BIM Electrical Technical Office Engineer

MTEC | Abu Dhabi, UAE | Jul 2025 – Present

- Produce shop drawings for electrical systems including power distribution, lighting and cable containment on Manchester City Yas Residences (Yas Island), Elie Saab Waterfront (Al Reem Island) and Jacob & Co. Beachfront Living (Al Jurf), all by Ohana.
- Coordinate with consultants and clients to obtain design approvals and resolve technical queries.
- Prepare technical submittals, load distribution schedules and compliance reports.
- Provide ongoing support to site teams for design clarifications and drawing revisions.

BIM Electrical Technical Office Engineer

C&P | Cairo, Egypt | May 2025 – Jul 2025

- Prepared electrical shop drawings for lighting, mechanical power, small power and light current systems.
- Assisted in preparing material submittals and coordinated with suppliers for approvals.
- Supported project BOQ preparation.

BIM Electrical Design Engineer

EGEC | Cairo, Egypt | Jul 2024 – May 2025

- Designed complete electrical systems for school projects under the Royal Commission for AlUla (RCU) using Revit and BIM standards.
- Delivered electrical design packages for Modon projects, covering power, lighting and LV distribution.
- Collaborated with mechanical and civil teams to identify and resolve model clashes in Navisworks.
- Prepared BOQs and construction documentation to support project delivery and contractor tendering.

BIM Electrical Technical Office Engineer

Albawani | Riyadh, Saudi Arabia | May 2023 – Jul 2024

- Developed BIM models and construction drawings for the Madina Entertainment Complex.
- Coordinated with architects and contractors on King Salman Park and the Royal Commission for AlUla projects.
- Conducted quantity take-offs and maintained accurate project documentation throughout design stages.

BIM Electrical Technical Office Engineer

Solid Construction | Cairo, Egypt | Dec 2021 – May 2023

- Produced shop drawings for IL-Bosco (New Administrative Capital) and Odyssia City (Mostakbal City).
- Optimized electrical layouts for the District 5 Marakez development to minimize rework during construction.
- Coordinated with site teams to issue design revisions and respond to RFIs within project timelines.

Electrical Technical Office Engineer

Power Tech Co. For Earthing | Cairo, Egypt | Dec 2020 – Dec 2021

- Prepared tender documents and BOQs for earthing and lightning protection systems.
- Produced technical drawings and specifications for earthing installations.
- Supported site supervision activities and responded to technical queries during project execution.

Automation Tools (Revit Add-ins)

- **Electrical Auto Wiring (C#, Revit API):** automatically circuits devices and draws filleted wiring for the whole project or a selected circuit, converts runs into conduits, and includes an MCP bridge for AI-assisted circuiting. Saves about 80% of the task time.
- **Lines to Conduits (C#, Revit API):** converts detail-line and model-line runs into conduits and elbows, with elevation from slab thickness and cover, connected to existing back boxes.
- **Auto Dimension (C#, Revit API, WPF):** dimensions electrical devices to walls, curtain-wall mullions and columns in linked models, room-aware and chained. Saves about 80% of the task time.
- **Model-to-BOQ Add-in (pyRevit):** turns the electrical model into the client's BOQ and Material Submittal for Manchester City Yas Residences, with live conduit and cable tray quantities and one-click Excel exports.

Education

Bachelor of Electrical Engineering

South Valley University | Sep 2015 – Jul 2020

- Grade: Good
- Graduation Project: Power System Distribution and Light Current (Grade: Excellent)

Certifications

- MEP BIM Diploma – Kaitech Scholarship, 2021
  Covered: MEP BIM modeling (mechanical, electrical and family creation), BIM coordination and clash detection, Dynamo automation and 4D to 7D BIM simulations
- Power System Distribution – JELECOM, 2020
  Covered: lighting design, power distribution, equipment sizing, system calculations, earthing and lightning systems, and low current systems
- Mastering Revit API Course – Elzohry Academy, 2023
  Covered: C# programming, Revit API foundations, Revit element manipulation, discipline-specific automation, advanced UI and architecture, and deployment.
- PLC and Automation – ATS, 2020

Technical Skills

Electrical Design: LV/MV distribution, earthing and lightning protection systems, load schedules, SLDs, cable sizing, voltage drop calculations, short-circuit calculations, lighting calculations, light current / ELV systems, fire alarm (NFPA 72), ADDC/AADC EWR.

Technical Office & BIM: shop drawings, IFC drawings, material submittals, RFIs, BOQ, MEP coordination, clash detection.

Software: Revit, AutoCAD, Navisworks, Dynamo, ETAP, DIALux evo, BIM 360 and ACC.

Soft Skills: Teamwork, Problem-Solving, Creativity, Adaptability and Attention to Detail.

Languages

Arabic: Native

English: Professional